A Novel Resistive Pulse Sensing Using Tunable Micro-/Nanopores for Analyte Size Characterization

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Abstract

Micro- and nanomaterials show strong links between their size features and functional behavior, making accurate size analysis essential for interpreting their physiological conditions. Resistive pulse sensing (RPS), which operates on the Coulter principle, allows real-time, label-free detection of single particles by monitoring transient changes in ionic resistance within a fluidic channel. In this work, we introduce a straightforward and structurally adjustable RPS platform fabricated through conventional photolithography. Micro-/nanopores were created by trapping microparticles inside bridge channels using evaporation-driven transport, enabling precise tuning of pore geometry through control of channel height and particle diameter. The platform effectively detected particles and biomaterials over a broad range of sizes. Overall, this low-cost and accessible RPS platform enables high-sensitivity, real-time characterization of various micro- and nanomaterials and holds promise for future biological and diagnostic applications.

Keywords: ***Resistive pulse sensing, Micro-/Nanopore, Micro-/Nanofluidics, Fluidic sensors***

1. Introduction

Size of micro-/nanomaterials is essential for understanding their functional properties and physiological states. While conventional techniques such as DLS and NTA have been widely used, they suffer from limitations including complex sample preparation and limited resolution for heterogeneous mixtures [1]. Resistive pulse sensing (RPS) has emerged as a powerful alternative, offering high sensitivity and real-time analysis of low-concentration analytes. However, most existing RPS platforms rely on costly and inflexible nanofabrication processes [2]. To address these challenges, this study introduces a facile RPS platform fabricated solely using conventional photolithography, with micro-/nanopores formed via evaporation-driven particle positioning. The proposed approach enables high structural flexibility, simple fabrication, and tunable pore dimensions, allowing quantitative analysis of particles across a wide size range, including mixed particles and cell-derived materials. The platform demonstrated its potential as an accessible and versatile tool for micro-/nanomaterial characterization.

2. Device concept and working principle

The RPS device was fabricated using standard photolithography and soft lithography, in which a pore was formed inside a bridge channel via evaporation-driven positioning of size-matched polystyrene microparticles. The pore size could be precisely tuned down to the submicron scale by adjusting the bridge channel height to 10 to 3 μm and particle diameter, with the bridge length designed to maximize resistance changes during particle translocation (Figure 1a–b). During operation, analytes suspended in electrolyte migrated electrophoretically through the particle-based pore under an applied electric field, generating transient voltage signals proportional to resistance changes. For 10 μm-bridge channel height, the device produced clear resistive pulses with amplitudes of ~2 V and a signal-to-noise ratio of 8–11 using 1 μm particles as analytes (Figure 1c–d).

3. Size-dependent signal characterization and validation of the pore-based RPS platform

Electrical signal measurements obtained from devices with different pore sizes are shown in Figure 2. For each device, three different particle diameters were selected according to the bridge channel height of 3, 6, and 10 μm, which determines the resulting pore size. As presented, the results showed that the signal intensity and peak count increased systematically with particle diameter, demonstrating that each device could independently resolve and distinguish particles of different sizes (Figure 2a–c). Furthermore, particle sizes calculated from the obtained RPS signals were in good agreement with reference values and with results obtained from DLS measurements, confirming the quantitative reliability of the platform (Figure 2d). Based on these comparisons, the effective measurement resolution of the proposed photolithography- and evaporation-based RPS platform was estimated to be approximately 400 nm.

The ability of the proposed RPS platform to analyze heterogeneous samples was demonstrated using ternary particle mixtures and cell-derived materials (Figure 3). Distinct peak populations were clearly resolved in ternary mixtures through Gaussian Mixture Model (GMM)-based analysis, confirming that multiple size components within mixed samples can be independently identified (Figure 3a–c). Minor deviations in peak statistics were observed due to size-dependent translocation dynamics and partial signal overlap in the multi-pore configuration. For devices with bridge channel heights of 10 and 6 μm, the extracted peak positions showed good agreement with reference particle sizes, while reduced resolution in 3 μm devices led to peak splitting without significantly distorting overall size trends. Compared to DLS, the RPS measurements provided more detailed size and abundance information for heterogeneous particle mixtures with good reproducibility. Applying the same approach to

cell-derived materials further demonstrated enhanced resolution of dominant size components relative to DLS, particularly for analytes larger than ~400 nm, indicating that the platform is well suited for quantitative size characterization of microvesicle-scale biological materials without additional labeling or preprocessing (Figure 3d–f).

Acknowledgments

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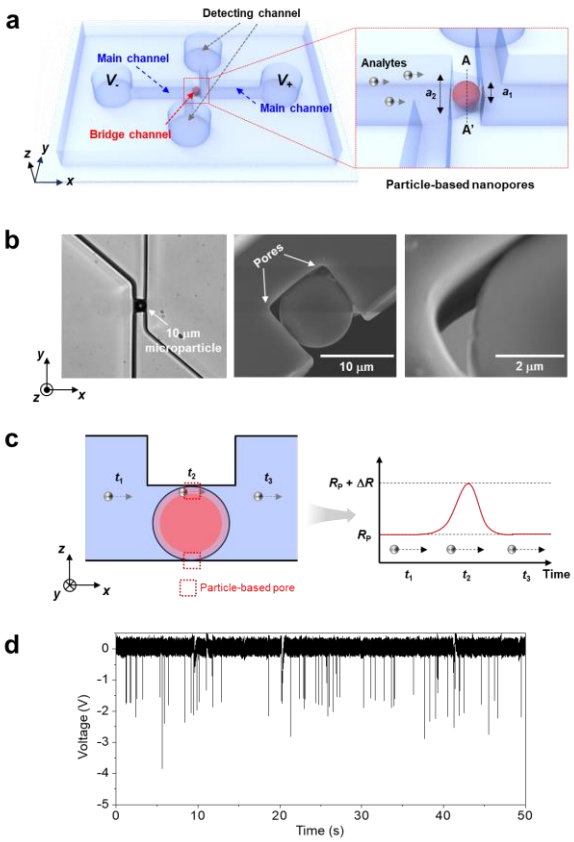

Fig. 1. Concept and working principle of the RPS platform. (a) Schematic of a RPS platform integrated with particle-based micro-/nanopores. (b) Representative optical and SEM images of the fabricated platform and the pore using 10 µm-high bridge channel. (c) Mechanism of electric signal generation induced by analyte translocation. (d) Representative electric signal measurement result.

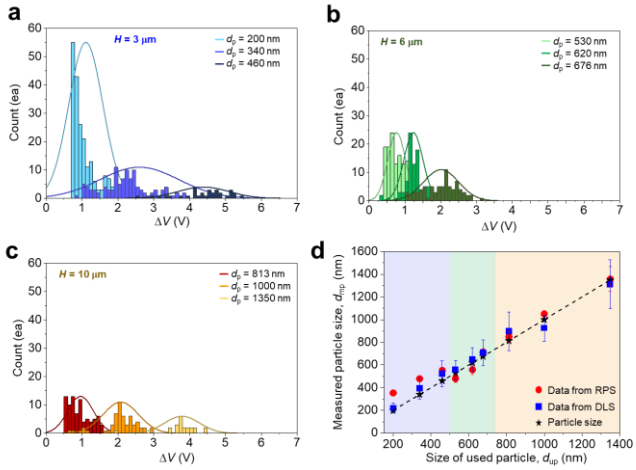

Fig. 2. Signal characterization of the RPS platform under different particle sizes. (a)–(c) Representative distributions of voltage change (ΔV) as a function of particle diameter using the platform with bridge channels of 3, 6, and 10-µm height, respectively. (d) RPS and DLS measurement results for particle diameters with respect to reference particle diameters.

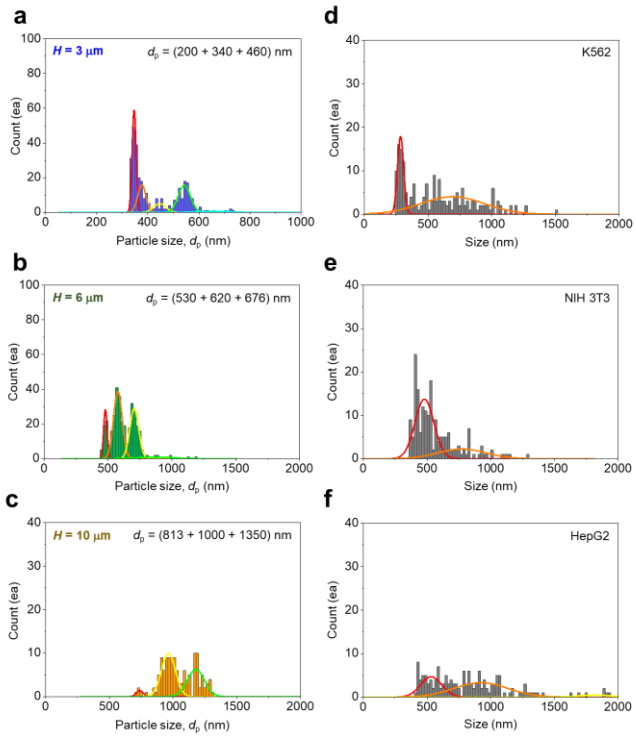

Fig. 3. RPS-based peak detection for the ternary particle mixtures and cell-derived materials. (a)–(c) Representative RPS measurement results for ternary particle mixtures containing particles with diameters of 200, 340, and 460 nm; 530, 620, and 676 nm; and 813 nm, 1.0 µm, and 1.35 µm, respectively. (d)–(f) Representative RPS measurement results for cell-derived materials, obtained from K562 (human leukemia), NIH 3T3 (mouse fibroblast), and HepG2 (human hepatocellular carcinoma) cells, respectively.